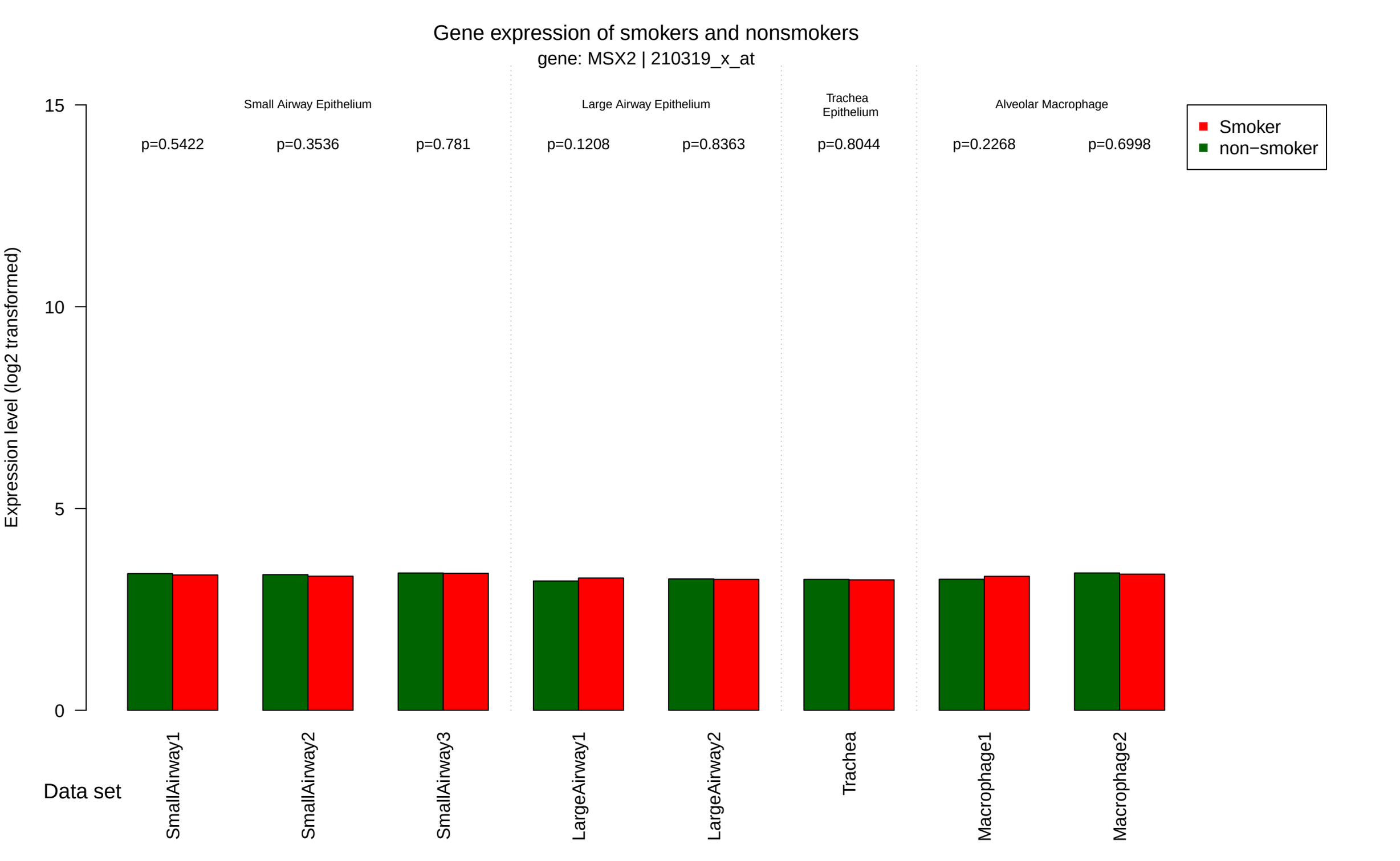
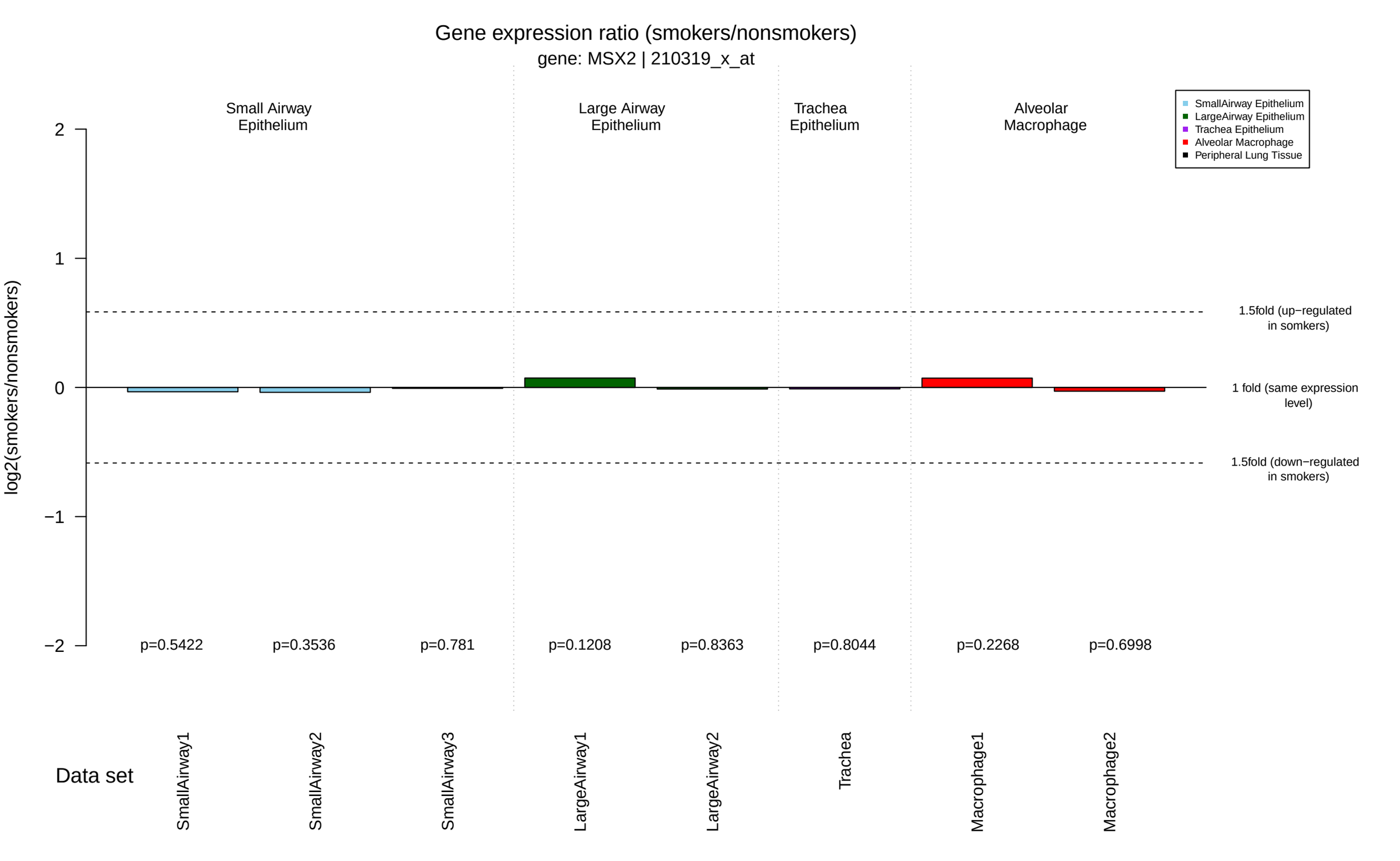
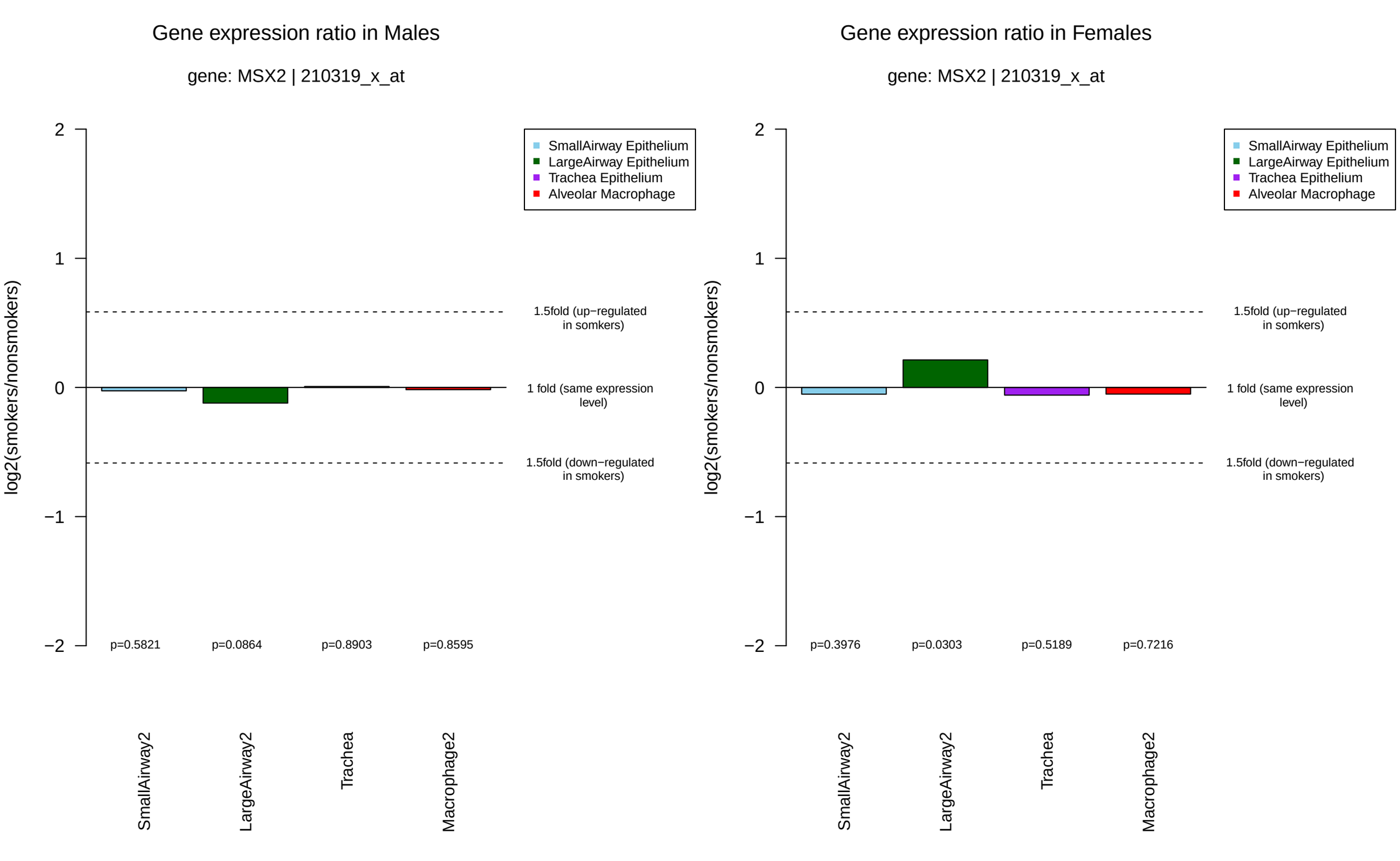
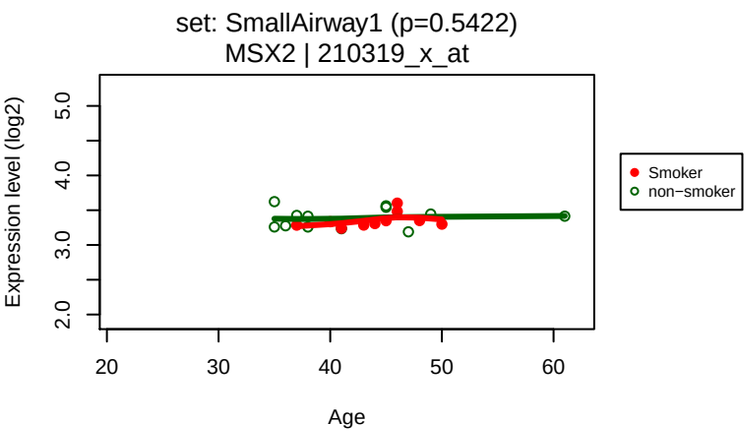
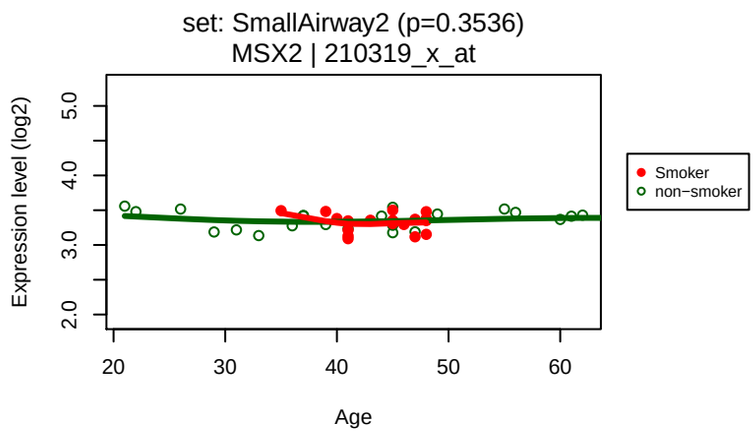
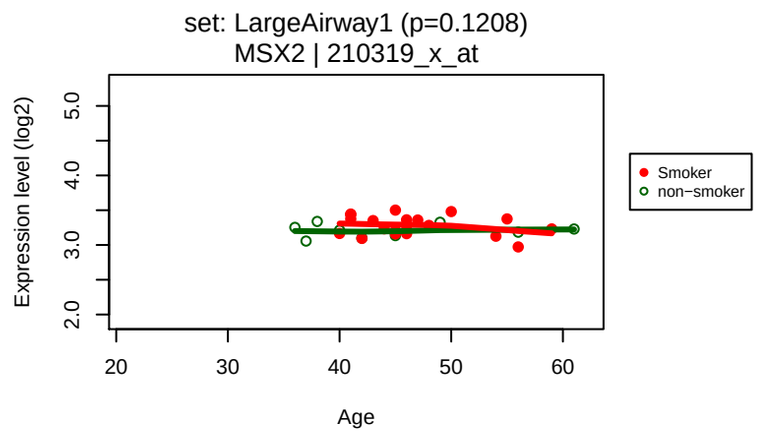

Gene expression of smokers and nonsmokers
gene: MSX2 | 210319_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5422
p=0.3536
p=0.781
p=0.1208
p=0.8363
p=0.8044
p=0.2268
p=0.6998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

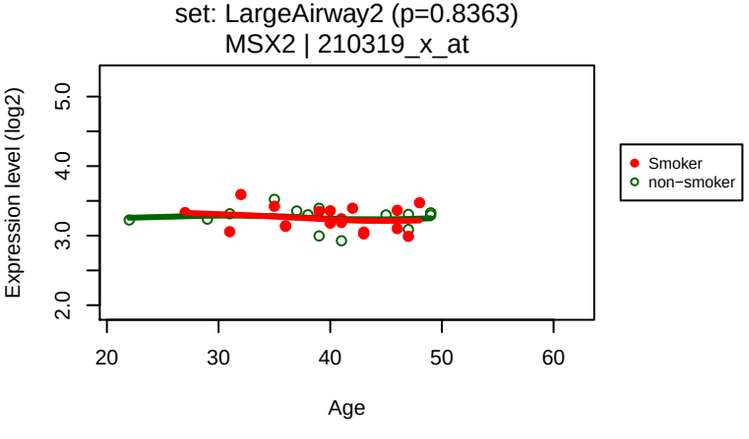

Gene expression ratio (smokers/nonsmokers)
gene: MSX2 | 210319_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5422
p=0.3536
p=0.781
p=0.1208
p=0.8363
p=0.8044
p=0.2268
p=0.6998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

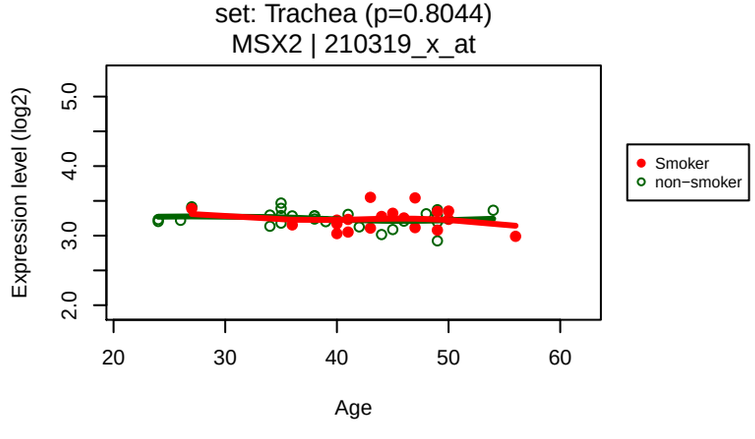

Gene expression ratio in Males
gene: MSX2 | 210319_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5821
p=0.0864
p=0.8903
p=0.8595
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MSX2 | 210319_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3976
p=0.0303
p=0.5189
p=0.7216
SmallAirway2
LargeAirway2
Trachea
Macrophage2

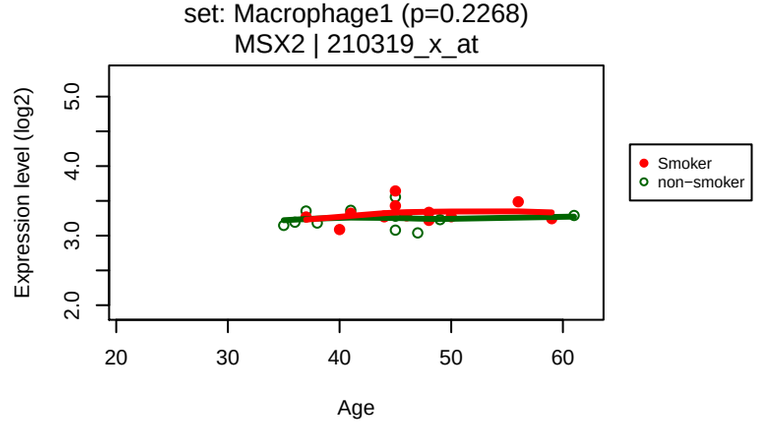

set: SmallAirway1 (p=0.5422)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker

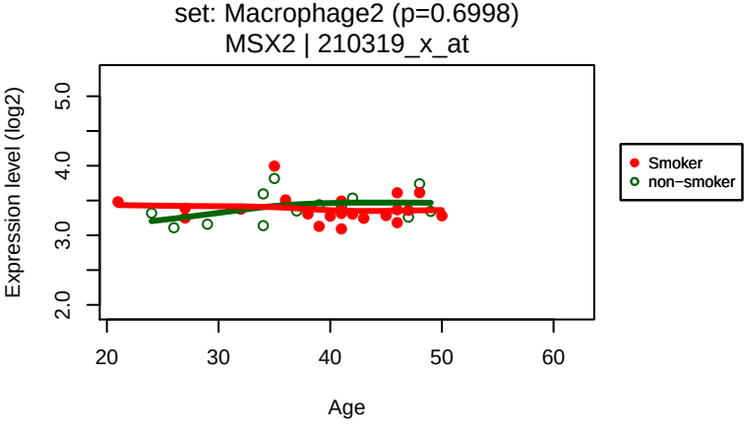

set: SmallAirway2 (p=0.3536)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker

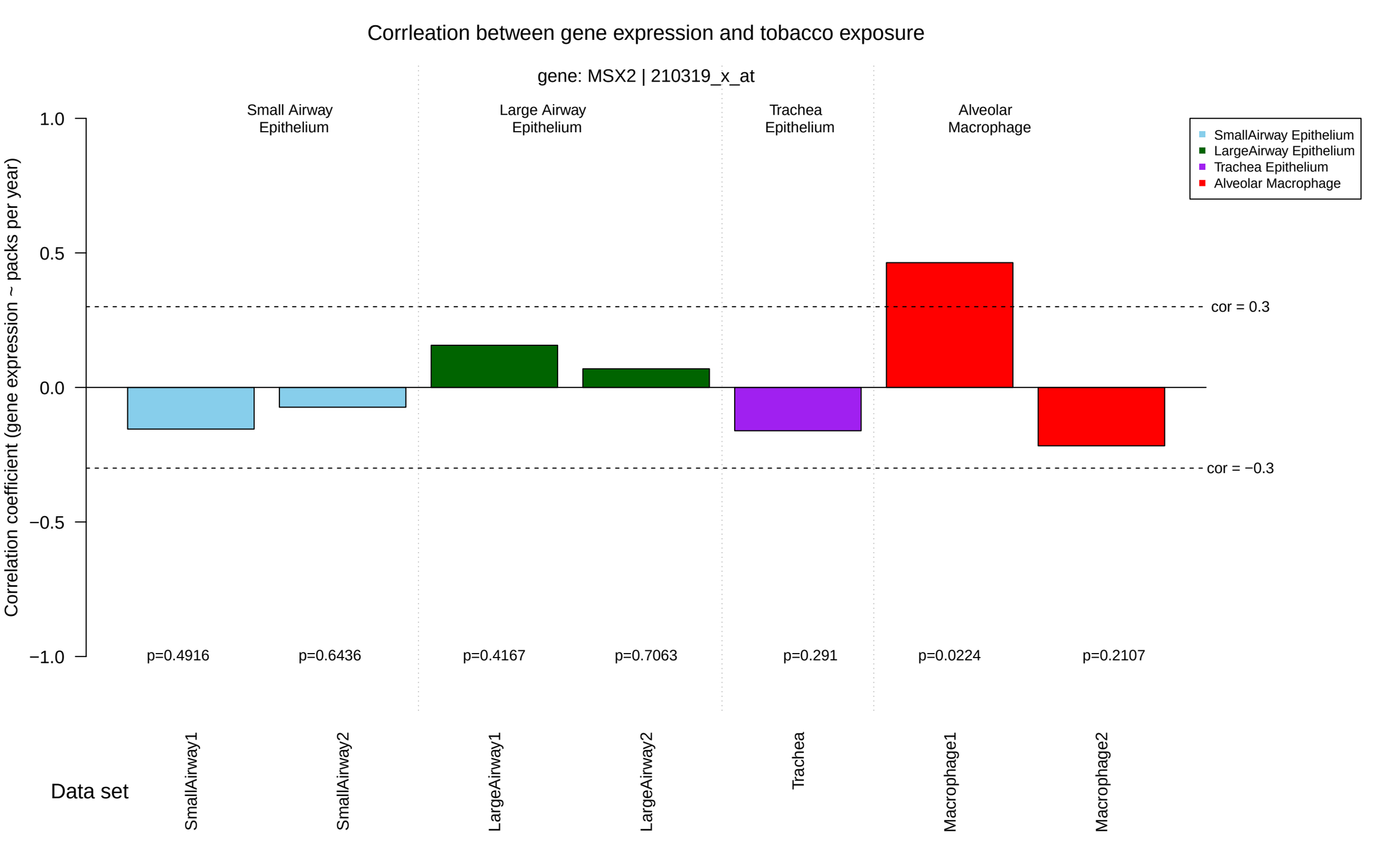

set: LargeAirway1 (p=0.1208)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8363)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8044)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2268)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6998)
MSX2 | 210319_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MSX2 | 210319_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4916
p=0.6436
p=0.4167
p=0.7063
p=0.291
p=0.0224
p=0.2107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2